



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

SENIOR CERTIFICATE EXAMINATIONS/
NATIONAL SENIOR CERTIFICATE EXAMINATIONS

ENGINEERING GRAPHICS AND DESIGN P1

MAY/JUNE 2026

MARKS: 100

TIME: 3 hours

This question paper consists of 6 pages.

Barcode label

DO NOT FOLD THE QUESTION PAPER IN HALF.

Confidential

INSTRUCTIONS AND INFORMATION

1. This question paper consists of FOUR questions.
2. Answer ALL the questions.
3. ALL drawings are in first-angle orthographic projection, unless otherwise stated.
4. ALL drawings must be prepared using pencil and instruments, unless otherwise stated.
5. ALL answers must be drawn accurately and neatly.
6. ALL the questions must be answered on the QUESTION PAPER, as instructed.
7. ALL the pages, irrespective of whether the question was attempted or not, must be re-stapled in numerical sequence in the TOP LEFT-HAND CORNER ONLY.
8. Time management is essential in order to complete all the questions.
9. Print your examination number in the block provided on every page.
10. Any details or dimensions not given must be assumed in good proportion.

FOR OFFICIAL USE ONLY											
QUESTION	MARKS OBTAINED			$\frac{1}{2}$	SIGN	MODERATED			$\frac{1}{2}$	SIGN	RE-MARKING
1											
2											
3											
4											
TOTAL											
	2	0	0			2	0	0			2 0 0

FINAL CONVERTED MARK

100

CHECKED BY

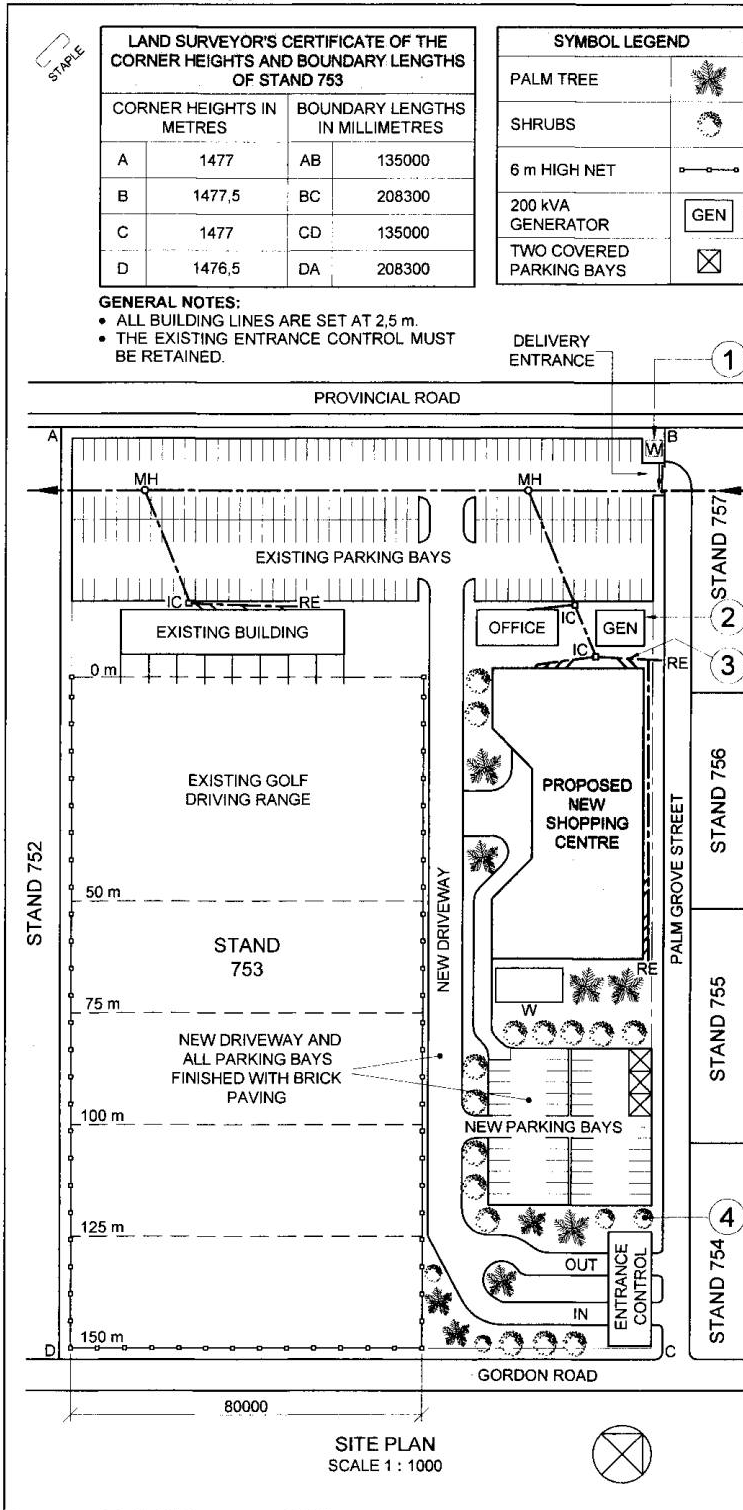
COMPLETE THE FOLLOWING:.

CENTRE NUMBER

CENTRE NUMBER

EXAMINATION NUMBER

EXAMINATION NUMBER



NOTE:
Contractors must verify all dimensions and levels on site before commencing work. Architects to be notified immediately of any discrepancies.

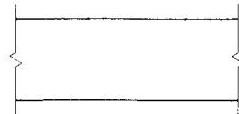
ARCHITECT'S SIGNATURE

CLIENT'S SIGNATURE

ANSWER 20

In the space below, draw, in neat freehand, the SANS 10143 graphical symbol for:
(a) FACE BRICK HATCHING
(b) The top view of a SHOWER

(a) FACE BRICK HATCHING



(b) SHOWER



1	14/05/2025	ADD 200 kVA GENERATOR
REVISION	DATE	DESCRIPTION
PRINTED BY:	DATE:	
OUR PRINTERS	11/07/2025	
ENGINEERING COMPANY:		
HAGE CIVIL ENGINEERS 222 TRICHARDT AVENUE VANDERBIJLPARK 1911 hage-civileng@gmail.com		
PROJECT ARCHITECT:		
E MOKOENA		
PROJECT TITLE:		
PROPOSED NEW SHOPPING CENTRE, DRIVEWAY AND PARKING BAYS FOR MR F DE JONGH, ON STAND 753, PALM GROVE STREET, VANDERBIJLPARK		
DRAWING TITLE:		
SITE PLAN		
PROJECT NUMBER:	DRAWING NUMBER:	
7530701-24	S P - 2 4 A 1 -	
REFERENCE CODE:	SCALE:	
04050124	1 : 1000	
DRAWN:	DATE:	CHECKED:
HENNIE	05/05/2025	GOUTA
		05/06/2025

QUESTION 1: ANALYTICAL (CIVIL)

Given:

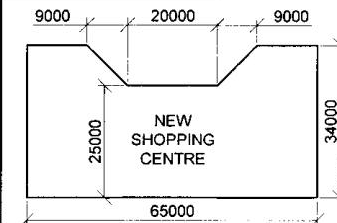
The site plan of an existing golf driving range with a proposed new shopping centre, driveway and parking bays, a title panel and a table of questions. The drawing is not presented to the indicated scale.

Instructions:

Complete the table below by neatly answering the questions, which refer to the accompanying drawing, title panel and civil content. Abbreviations may not be used, unless otherwise stated. [30]

QUESTIONS		ANSWERS	
1	What scale is indicated for the site plan?	1	
2	Who checked the drawing?	1	
3	What is the project architect's name?	1	
4	What must the contractors do on site before they commence with the work?	1	
5	In what street is the delivery entrance?	1	
6	How many palm trees are shown on the site plan?	1	
7	What is the distance from the building lines to the boundary lines on STAND 753 in millimetres?	1	
8	Name the feature at 1.	1	
9	What is the kVA capacity of the feature at 2?	1	
10	What feature has not been labelled on the sewerage line at 3?	1	
11	Name the feature at 4.	1	
12	How many covered parking bays are there on STAND 753?	1	
13	Which corner of STAND 753 is the highest above sea level?	1	
14	What is the finish on the new driveway and parking bays?	1	
15	What does the abbreviation IC on a sewerage line stand for?	2	
16	What is indicated in GREEN on elevations?	2	
17	With reference to the north point, what is the direction of flow in the municipal sewerage line?	2	
18	In the space below (ANSWER 18), determine the total length of the 6 m high net.	3	
19	In the space below (ANSWER 19), determine the total area of the proposed new shopping centre in square metres.	3	
20	In the space in the title panel (ANSWER 20), draw, in neat freehand, the SANS 10143 graphical symbol for: (a) FACE BRICK HATCHING and (b) the top view of a SHOWER.	4	
TOTAL		30	

ANSWER 18
Show ALL calculations.



ANSWER 19
Show ALL calculations.

EXAMINATION NUMBER	
EXAMINATION NUMBER	2

STAPLE

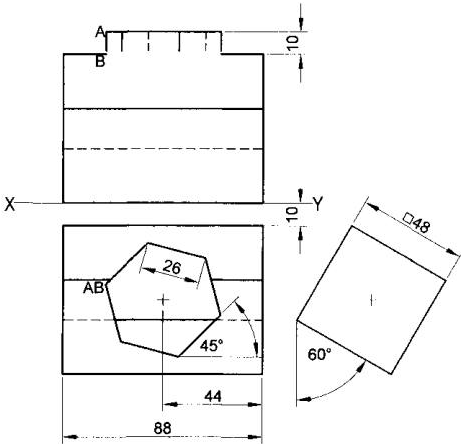
QUESTION 2: INTERPENETRATION AND DEVELOPMENT

- Given:**
- The top view and the incomplete front view of a right regular hexagonal tube that has been shaped to join onto a right square tube. The axes of both tubes lie in a common vertical plane.
 - An auxiliary view of the square tube

- Instructions:**
- Draw, to scale 1 : 1, the following views of the TWO tubes:
- 2.1 The given top view
 - 2.2 A right view
 - 2.3 The complete front view, clearly showing the curve of interpenetration
 - 2.4 Develop the hexagonal tube. Make edge 'AB' the seam.

- Planning is essential.
- Show ALL construction.
- Show ALL hidden detail.

[38]



ASSESSMENT CRITERIA				
1	TOP VIEW	10 ½		
2	RIGHT VIEW	7 ½		
3	FRONT VIEW	11 ½		
4	DEVELOPMENT	8 ½		
PENALTIES (-)				
TOTAL		38		
EXAMINATION NUMBER				
EXAMINATION NUMBER				3

STABLE

QUESTION 3: PERSPECTIVE**Given:**

Three views of an outdoor pottery studio and the information needed to draw a two-point perspective drawing

PP – Picture plane

HL – Horizon line

GL – Ground line

SP – Station point

Instructions:

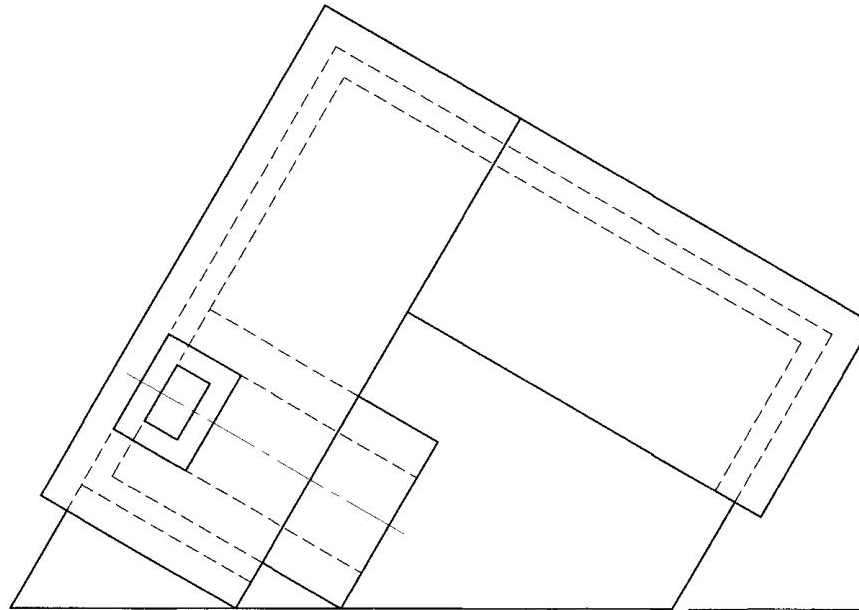
Complete the perspective drawing.

- Align the drawing sheet with the ground line (GL).
- Determine and label the vanishing points.
- Show ALL construction.
- Show ALL visible details.
- NO hidden detail is required.

[40]

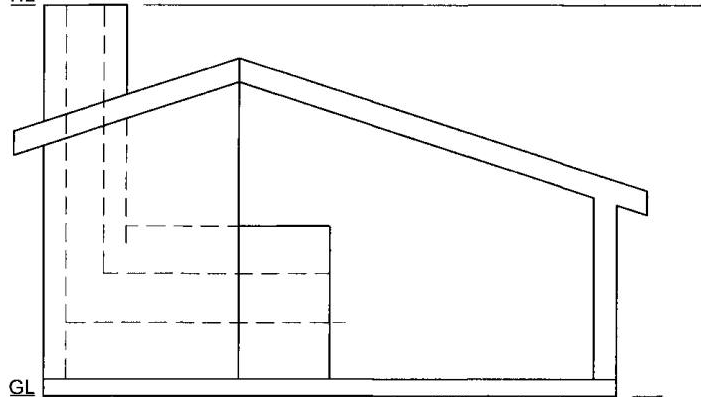
PP

PP



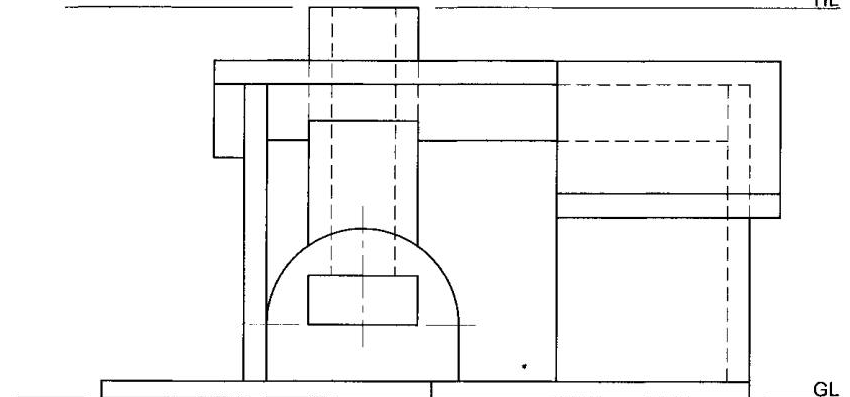
HL

HL



GL

GL

**ASSESSMENT CRITERIA**

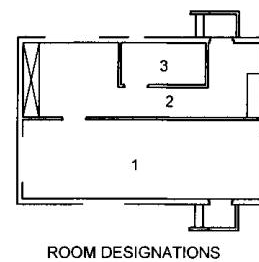
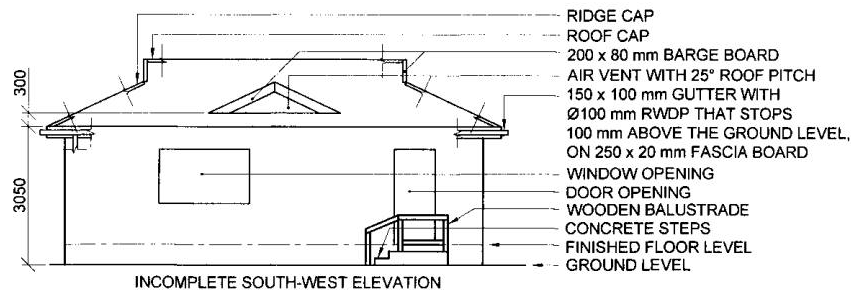
1	CONSTRUCTION	6			
2	FLOOR + WALLS	9 ½			
3	ROOF + CHIMNEY	12 ½			
4	OVEN	12			
PENALTIES (-)					
TOTAL		40			

SP

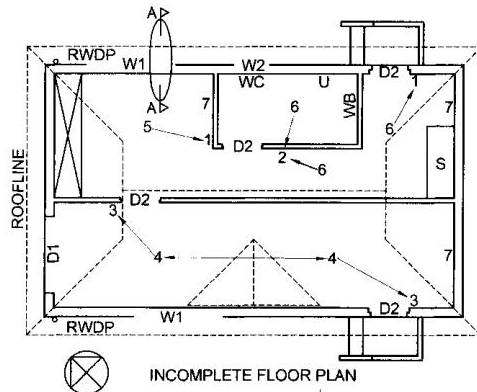
EXAMINATION NUMBER

EXAMINATION NUMBER

4

**FLOOR FINISHES**

1. WORKSHOP: EPOXY TILE
 2. OFFICE AND KITCHEN: TILE
 3. TOILET: TILE

**FEATURES**

- D1 ROLL-UP DOOR
 D2 DOOR
 W1 WINDOW
 W2 WINDOW

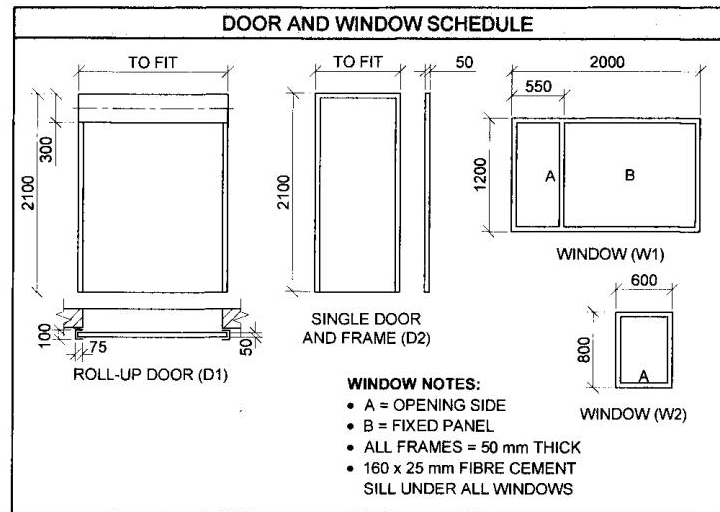
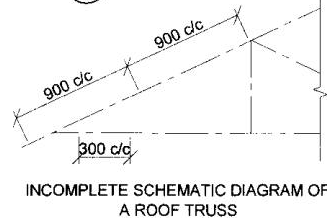
FIXTURES

- S SINK
 WC TOILET
 WB WASHBASIN
 U WALL-MOUNTED URINAL

ELECTRICAL FITTINGS

1. ONE-WAY SWITCH - SINGLE-POLE
 2. ONE-WAY SWITCH - DOUBLE-POLE
 3. TWO-WAY SWITCH
 4. FLUORESCENT LIGHT 4 x 40 W
 5. FLUORESCENT LIGHT 2 x 40 W
 6. CEILING LIGHT
 7. SWITCHED SOCKET OUTLET

NOTE:
 THE ARROW SHOWS THE LIGHT CONNECTION TO THE SWITCH.

**WINDOW NOTES:**

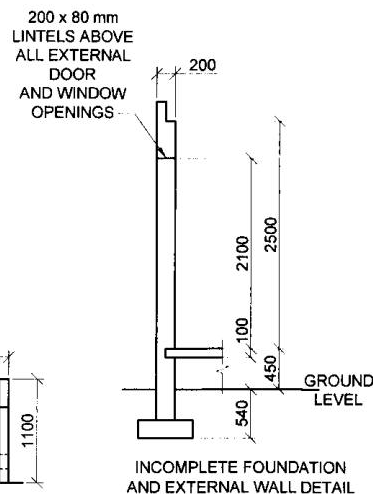
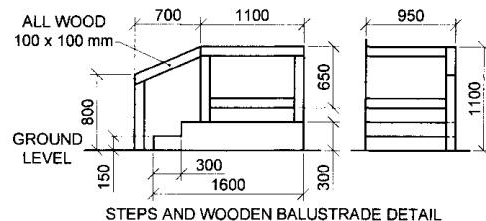
- A = OPENING SIDE
- B = FIXED PANEL
- ALL FRAMES = 50 mm THICK
- 160 x 25 mm FIBRE CEMENT SILL UNDER ALL WINDOWS

ROOF COMPONENTS		ELECTRICAL SYMBOLS	
	ROOF CAP AND RIDGE CAP		
	200 x 80 mm BARGE BOARD		
	250 x 20 mm FASCIA BOARD		
	150 x 100 mm GUTTER		
	80 x 50 mm PURLIN		

FIXTURE SCHEDULE			
TOILET (WC)	SINK (S)	WALL-MOUNTED URINAL (U)	WASHBASIN (WB)

ROOF NOTES:

- 25° ROOF PITCH
- 120 x 40 mm ROOF TRUSSES ON
- 120 x 40 mm WALL PLATES
- 400 mm ROOF OVERHANG TO END OF ROOF TRUSS
- 40 mm CORRUGATED ROOF SHEET ON 80 x 50 mm PURLINS @ 900 mm c/c
- 200 x 80 mm BARGE BOARDS ON GABLE ENDS
- 250 x 20 mm FASCIA BOARDS WITH 150 x 100 mm GUTTERS ON ALL SIDES
- 10 mm CEILING BOARDS ON 40 x 40 mm BRANDING STRIPS @ 300 mm c/c

**QUESTION 4: CIVIL DRAWING****Given:**

- The incomplete south-west elevation of a **new workshop**, showing the walls, the window and door openings, steps, wooden balustrade, roof and labels
- The incomplete floor plan showing the walls, roof lines, steps, positions of the doors, windows, fixtures and electrical layout
- An incomplete schematic diagram of a roof truss
- The steps and wooden balustrade detail
- The incomplete foundation and external wall detail
- Room designations and floor finishes
- A door and window schedule
- A table of roof components
- A table of electrical symbols
- A fixture schedule
- The incomplete floor plan and position of the ground level of the **new workshop**, drawn to scale 1 : 50, and the incomplete foundation and break line for the detailed section, drawn to scale 1 : 20, on page 6

Instructions:

Answer this question on page 6.

- 4.1 Using the given incomplete floor plan and ground level, draw, to scale 1 : 50, the following views of the **new workshop**:

4.1.1 THE COMPLETE FLOOR PLAN

Add the following features to the drawing:

- ALL doors and windows
- ALL fixtures as indicated by the abbreviations
- ALL electrical fittings as indicated by the numbers
- ALL hatching detail

4.1.2 THE COMPLETE SOUTH-WEST ELEVATION

Show the following features on the drawing:

- The wall, window, door, step and wooden balustrade detail
- The roof detail including air vent, barge boards, fascia board, gutter and rainwater downpipe
- The finished floor level

- 4.2 Using the incomplete foundation and break lines on page 6, draw, to scale 1 : 20, a **DETAILED SECTION** on cutting plane A-A of the area in the ellipse shown on the incomplete floor plan.

Show the following features on the drawing:

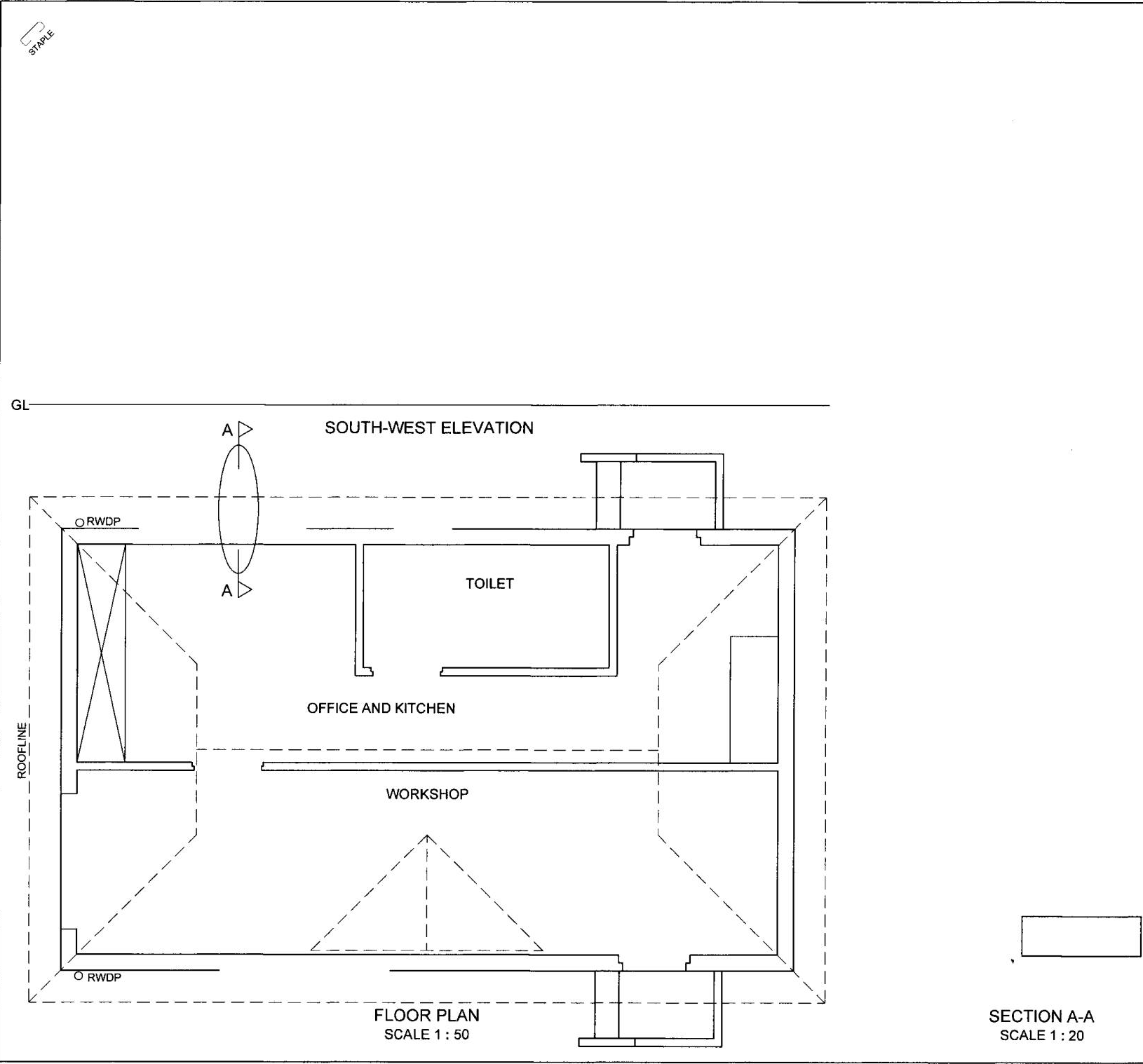
- The complete foundation, external wall, slab and window detail
- The roof detail, including the fascia board and gutter
- The steps and balustrade detail to the right (south-east) of cutting plane A-A
- ALL hatching detail. ONLY the substructure hatching may be drawn in neat freehand.

Label the following:

- The floor finishes
- Ground level, finished floor level and damp-proof course (use the correct abbreviations and show them on ALL relevant views)

NOTE:

ALL drawings must comply with the **guidelines** and **graphical symbols** as contained in **SANS 10143**. [92]



MARK ALLOCATION FOR SECTION OF ROOF			FOR OFFICIAL USE ONLY			
A			INCORRECT SCALE(S) USED			
B			NON-ALIGNMENT OF VIEWS			
C			VIEW(S) ROTATED			
D			SECTION VIEWED INCORRECTLY			
E			INCORRECT LETTERING			
F						
G						
H						
J						
TOTAL			TOTAL			

ASSESSMENT CRITERIA					
FLOOR PLAN					
		POSSIBLE	OBTAINED	SIGN	MODERATED
1	DOORS + WINDOWS	11 ½			
2	FIXTURES	7 ½			
3	ELECTRICAL	12			
4	LABELS	1 ½			
5	HATCHING	3			
SUBTOTAL		35 ½			
SOUTH-WEST ELEVATION					
1	ROOF + RWDP	11			
2	WALLS + BALUSTRADE	7 ½			
3	DOORS + WINDOWS	6 ½			
4	LABELS	½			
SUBTOTAL		25 ½			
DETAILED SECTION					
1	ROOF DETAIL	11			
2	SLAB + WALL	8 ½			
3	WINDOW	4 ½			
4	HATCHING	5 ½			
5	LABELS	1 ½			
SUBTOTAL		31			
TOTAL		92			
PENALTIES (-)					
GRAND TOTAL					
EXAMINATION NUMBER					
EXAMINATION NUMBER					6